```
import pandas as pd
import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt

from sklearn.model_selection import train_test_split
from sklearn.metrics import confusion_matrix
from sklearn.metrics import accuracy_score,f1_score,precision_score,recall_score,r2_score
from sklearn.tree import DecisionTreeClassifier
```

```
df=pd.read_csv("/content/Admission_Predict (1).csv")
df.head()
```

	Serial No.	GRE Score	TOEFL Score	University Rating	SOP	LOR	CGPA	Research	Chance of Admit
0	1	337	118	4	4.5	4.5	9.65	1	0.92
1	2	324	107	4	4.0	4.5	8.87	1	0.76
2	3	316	104	3	3.0	3.5	8.00	1	0.72
3	4	322	110	3	3.5	2.5	8.67	1	0.80
4	5	314	103	2	2.0	3.0	8.21	0	0.65

Next steps: Generate code with df New interactive sheet

```
df.shape
```

```
(400, 9)
```

```
df.describe()
```

	Serial No.	GRE Score	TOEFL Score	University Rating	SOP	LOR	CGPA	Research	Chance of Admit
count	400.000000	400.000000	400.000000	400.000000	400.000000	400.000000	400.000000	400.000000	400.000000
mean	200.500000	316.807500	107.410000	3.087500	3.400000	3.452500	8.598925	0.547500	0.724350
std	115.614301	11.473646	6.069514	1.143728	1.006869	0.898478	0.596317	0.498362	0.142609
min	1.000000	290.000000	92.000000	1.000000	1.000000	1.000000	6.800000	0.000000	0.340000
25%	100.750000	308.000000	103.000000	2.000000	2.500000	3.000000	8.170000	0.000000	0.640000
50%	200.500000	317.000000	107.000000	3.000000	3.500000	3.500000	8.610000	1.000000	0.730000
75%	300.250000	325.000000	112.000000	4.000000	4.000000	4.000000	9.062500	1.000000	0.830000

```
df.dtypes
```

	0
Serial No.	int64
GRE Score	int64
TOEFL Score	int64
University Rating	int64
SOP	float64
LOR	float64
CGPA	float64
Research	int64
Chance of Admit	float64

dtype: object

```
df.isnull().sum()
```

	0
Serial No.	0
GRE Score	0
TOEFL Score	0
University Rating	0
SOP	0
LOR	0
CGPA	0
Research	0
Chance of Admit	0

dtype: int64

```
df = df.dropna()
```

```
df['Admitted']=df['Chance of Admit ']>=0.75
df['Admitted']=df['Admitted'].astype(int)
df=df.drop('Chance of Admit ',axis=1)
```

```
df.head()
```

	Serial No.	GRE Score	TOEFL Score	University Rating	SOP	LOR	CGPA	Research	Admitted
0	1	337	118	4	4.5	4.5	9.65	1	1
1	2	324	107	4	4.0	4.5	8.87	1	1
2	3	316	104	3	3.0	3.5	8.00	1	0
3	4	322	110	3	3.5	2.5	8.67	1	1
4	5	314	103	2	2.0	3.0	8.21	0	0

Next steps: Generate code with df New interactive sheet

```
x = df[['GRE Score','TOEFL Score','University Rating','SOP','CGPA','Research']]
```

```
y=df[['Admitted']]
```

```
x_train,x_test,y_train,y_test=train_test_split(x,y,test_size=0.2,random_state=42)
```

```
dt = DecisionTreeClassifier(max_depth=3)
dt.fit(x_train,y_train)
```

```
DecisionTreeClassifier
DecisionTreeClassifier(max_depth=3)
```

```
y_pred=dt.predict(x_test)
```

```
acc=accuracy_score(y_test,y_pred)
precision=precision_score(y_test,y_pred)
recall=recall_score(y_test,y_pred)
f1=f1_score(y_test,y_pred)
print(acc)
print(precision)
print(recall)
print(f1)
```

```
0.95
0.9393939393939394
0.9393939393939394
0.9393939393939394
```

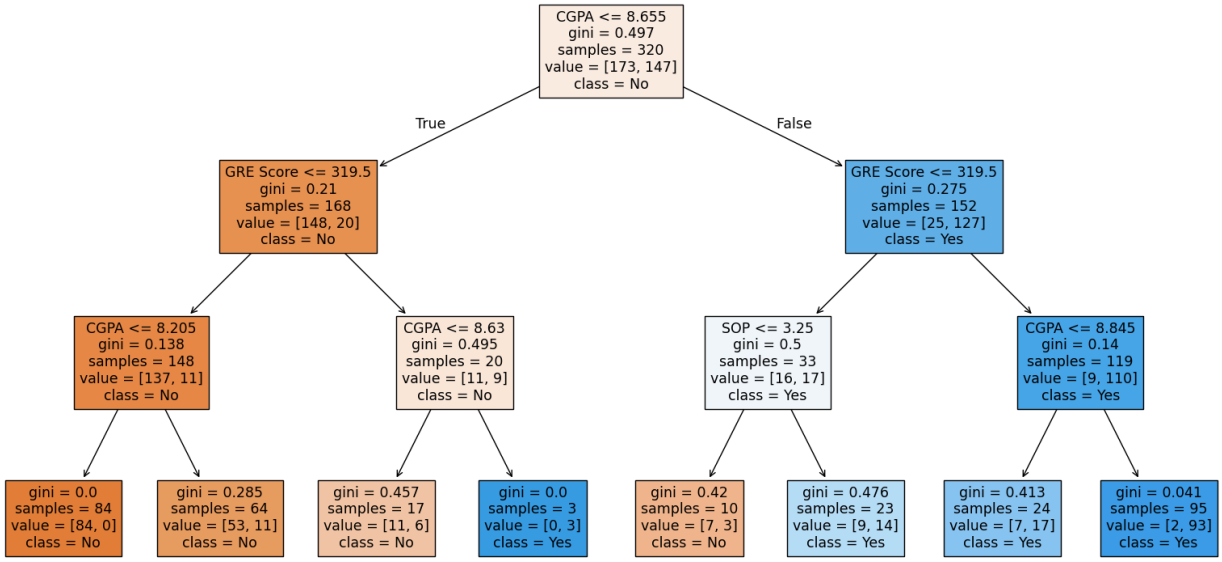
```
cm=confusion_matrix(y_test,y_pred)
cm
```

```
array([[45,  2],
       [ 2, 31]])
```

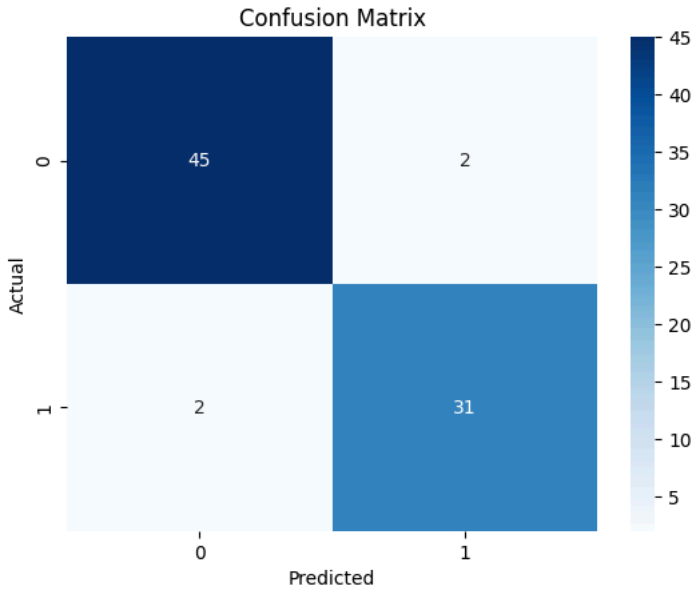
```
train_pred = dt.predict(x_train)

print("Train Accuracy =", accuracy_score(y_train , train_pred))
print("Test Accuracy =", accuracy_score(y_test , y_pred))
```

```
Train Accuracy = 0.88125
Test Accuracy = 0.95
```

```
from sklearn.tree import plot_tree
plt.figure(figsize=(20,10))

plot_tree(dt,
          feature_names=x.columns,
          class_names=['No','Yes'],
          filled=True)

plt.show()
```




```
import seaborn as sns

sns.heatmap(cm,
            annot=True,
            fmt='d',
            cmap='Blues')

plt.xlabel("Predicted")
plt.ylabel("Actual")
plt.title("Confusion Matrix")
plt.show()
```


Confusion Matrix
45
2
2
31
Actual
0
1
Predicted
0
1
45
40
35
30
25
20
15
10
5